

# Scott J. Gilbert

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Aspiring computer engineer with a strong foundation in hardware design, embedded systems, and software development. Skilled in C, C++, Python, embedded systems, and data processing, with hands-on experience in real-time systems, machine learning, CI/CD, and web development.

## Education

### University of Illinois Urbana-Champaign

GPA: 4.00/4.00

**Activities and Societies:** Campus Honors Program, Illini Solar Car, Project: Code, Illini Redstone Computing, Engineers without Borders, Marching Illini  
**Relevant Coursework:** Calculus III (MATH 241), Intro Differential Equations (MATH 285), Intro to Computing (ECE 120), Computational Linear Algebra (MATH 257), Intro to Electronics (ECE 110), Computers & Programming (ECE 220), Quantum Physics (PHYS 214), Thermodynamics (PHYS 213)

### James B. Conant High School

Weighted GPA: 4.92/4.00 | Unweighted GPA: 4.00/4.00

**Activities:** Congressional Debate, Marching Band, Jazz Band, Honor Band/Orchestra, Cross Country, Track and Field, National Honor Society

**Relevant Coursework:** AP Calculus BC, AP Computer Science Principles, AP Computer Science A, AP Physics C

*B.S., Computer Engineering, May 2029*

*Grainger College of Engineering*

*High School Diploma, May 2025*

## Professional Experience

### Computer Administrator

October 2025 - Present

*University of Illinois Campus Honors Program (CHP)*

Urbana, Illinois

- Maintain CHP's computing hardware infrastructure, integrations with external storage and web services.
- Collaborate with university admissions, IT services, and human resources to process thousands of data points across pipelines in a timely manner.

### Full-Stack/AI Intern

June 2025 - August 2025

*Team2Go Inc.*

Yongin-si, Gyeonggi, South Korea (Remote)

- Engineered Dockerized systems for internal and external-facing tools, enhancing integration between backend applications and user interfaces.
- Integrated custom OpenAI GPT models with streaming output, document-based vector embedding, and rigid boundary/instruction declaration.
- Collaborated with executives through regular progress updates and delivered projects ahead of schedule while exploring LLM design/psychology.

## Leadership/Involvement

### Electrical Team/Array Team Member

Illini Solar Car | Oct 2025 - Present

- Develop maximum power point tracking firmware and solar array layout specifications, contributing to overall electrical and vehicle optimization.
- Participate in battery characterization, public showcases/vehicle demonstrations, composites fabrication, and other electrical/general car work.
- Hold core leadership positions as Webmaster, Array/MPPT lead, and lead radio technician for 2026-2027.

### President & Systems Administrator

Illini Redstone Computing | Jan 2026 - Present

- Manage day-to-day financial, administrative, and technical decision-making while acting as primary liaison to the university/other organizations.
- Develop and maintain a network of containerized applications across gaming, authentication, user management, and data pipelining services.

### Backend Team Lead

Project: Code UIUC | Sep 2025 - Dec 2025

- Managed team focused on deploying data handling, APIs, and core logic for an data-driven, all-in-one crop monitoring and plant health solution.
- Held weekly meetings, maintained production software, specified versioning and CI systems, track progress, and reported progress to project lead.

### Eagle Scout

Scouting America | Mar 2018 - Dec 2024

- Planned and organized a coordinated trail restoration effort where volunteers contributed over 100 service hours.
- Collaborated with sponsors to secure necessary resources/tooling to finish the project a week ahead of schedule with zero workplace incidents.

## Projects

### Personal Content System

May 2025 - Present

- Developing flexible microservices-based architectures integrating a portfolio, blog, authentication, account management, AI-driven content development, workflow automation & CI/CD tooling and service integration, and a rich, data-driven backend through container orchestration.
- Modernizing backend with scalable technologies like GraphQL for improving performance and expanding application and cross-platform use cases.

### Illinois Congressional Debate Association Website

May 2024 - August 2025

- Led development of a full-stack application, improving front-end utility with React and performance/scalability with a backend service and CMS.
- Programmed, tested, and deployed features improving mobile accessibility, information flow, site security, and page/component load times.
- Achieved strong search visibility, reaching ~20K impressions/6% CTR over four ranking among Google's top ten student congress sites.

### Solar Energy Educational Demonstration

October 2024 - April 2025

- Developed Arduino-based systems for temperature data acquisition, with custom firmware to interface with real-time visualization tools.
- Created interactive demonstrations, integrated computing elements, and technical infographics to teach solar concepts and energy efficiency.
- Collaborated with event organizers to deploy demonstrations for children and families at a community Earth Day event.

## Skills

- **Programming:** C, C++, Python, SQL, Java, Assembly, TypeScript, PHP, Wolfram
- **Databases:** MySQL, SQLite, PostgreSQL, pgvector, S3, Uploadthing, Backblaze
- **Numerical Analysis:** Jupyter Notebook, NumPy, SciPy, Matplotlib, Mathematica, Excel, Google Sheets
- **Embedded Systems/Microcontrollers:** Arduino, ELEGOO, KiCad, NXP
- **Industry Tools:** Linux, Docker, Docker Compose, Xilinx Vivado, MCUxpresso, LTSpice, Git, GitHub, Google Workspace, Microsoft Office, Slack
- **Workflow Automation & CI/CD:** Vercel, Claude Code, GitHub Copilot, GitHub Actions, n8n, Make
- **Technical Expertise:** Firmware Design, Backend Development, PCB Design, AI/ML Application, Virtualization, Object-Oriented Programming

**Awards:** Dean's List, National Merit Scholarship, AP Scholar with Distinction, Illinois State Seal of Biliteracy, Gold President's Volunteer Service Award